



ML FOR TITANIUM CARBIDE MXene CANCER SENSORS

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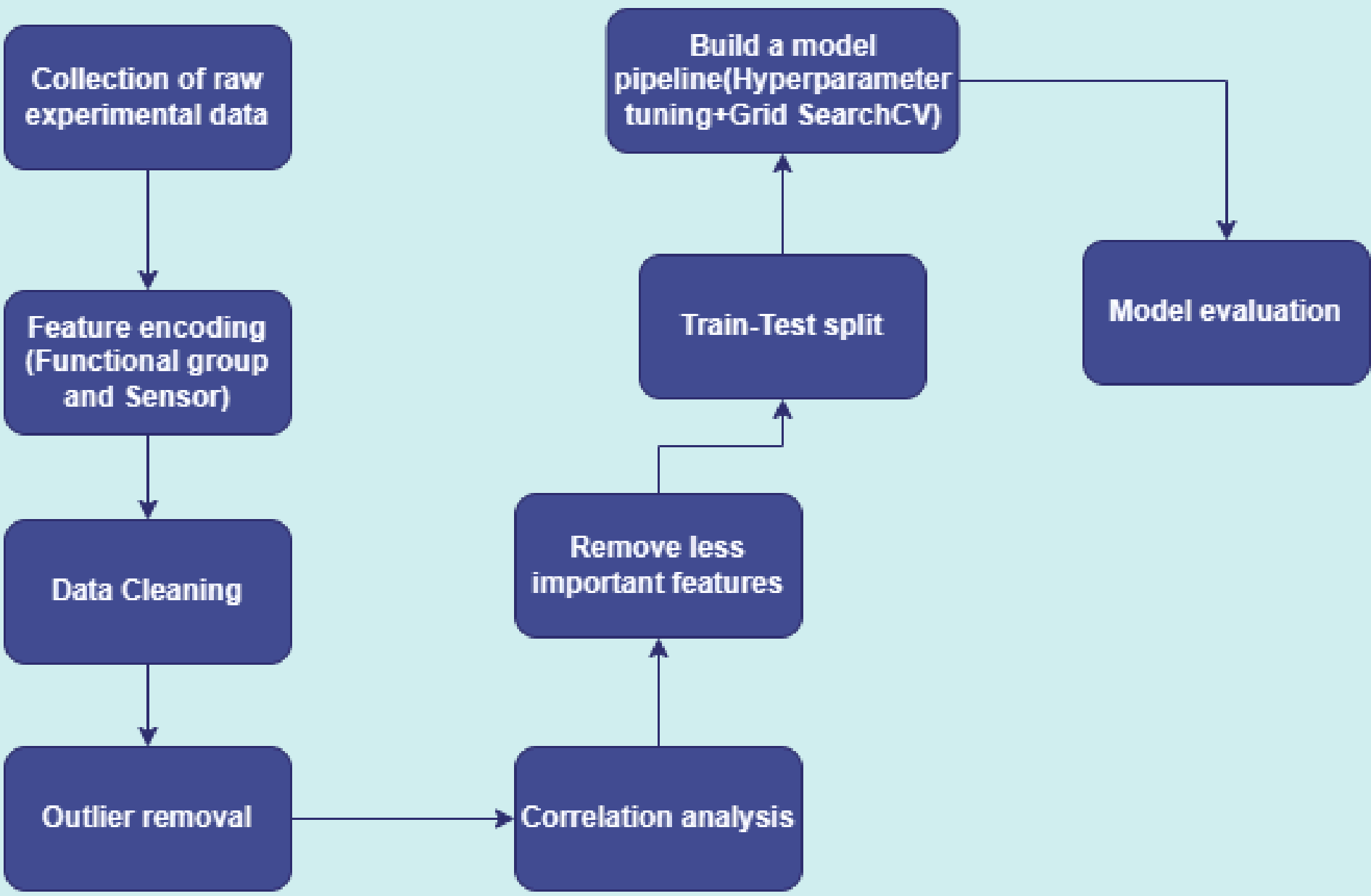
AIM

To study the effect of functional groups present on the Titanium carbide sensors on the detection of analytes, using a machine-learning based approach

Introduction

- Cancer is a leading cause of death worldwide; early detection through cancer biomarkers is crucial for improving survival rates and monitoring disease progression.
- Electrochemical biosensors offer high sensitivity, rapid response, portability, and cost-effectiveness, making them promising tools for biomarker detection.
- Ti_3C_2Tx MXenes enhance sensor performance due to their high conductivity and tunable surface functional groups ($-OH$, $-O$, $-F$), which strongly influence sensing behavior.
- This study applies a machine learning approach to analyze the impact of these functional groups, aiming to optimize sensor design for improved sensitivity, selectivity, and early cancer detection.

Methodology



Tools Used

1. Python
2. Pandas
3. Numpy
4. Seaborn
5. Scikit-learn
6. Jupyter notebook

Input features

1. Sensor_encoded
2. C/H ratio of analyte
3. C/O ratio of analyte
4. Properties of functional groups like EN, Atomic polarizability etc.
5. Functional group encoded
6. Interaction feature (Polarizability x C/H)

Data collection info

1. Dataset size : 137 rows x 16 columns
2. Source of data collection - Literature papers that focus on Ti_3C_2 MXene sensors. Journal links given in the google sheet link attached
3. Date Range : 2017-2026
4. Train-Test split : 80-20 split

ML models trained

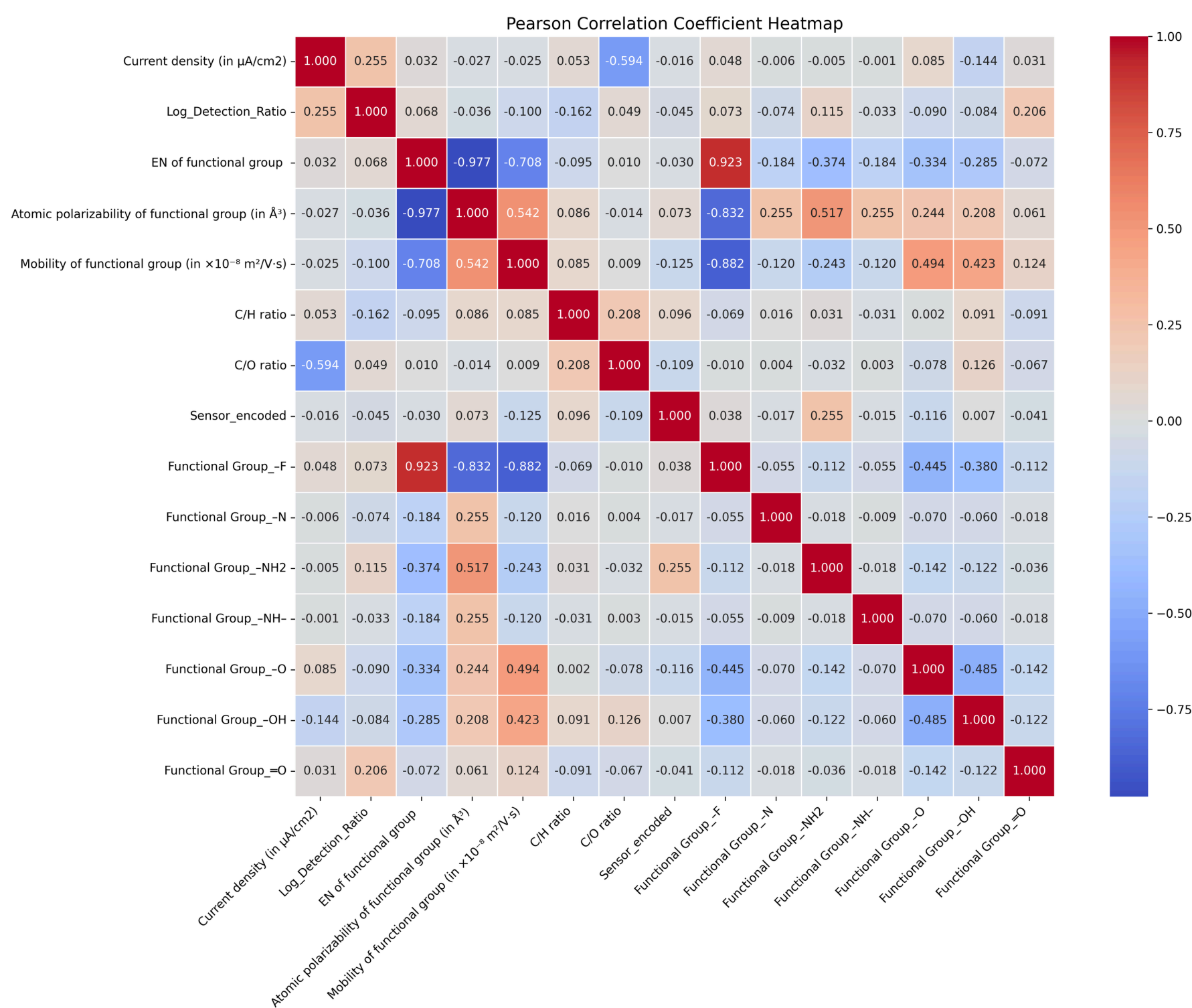
1. Gradient Boosting Regressor (both single and multi output)
2. Random Forest Regressor (both single and multi output)

Target outputs

1. Log Detection Ratio
2. Current Density

RESULTS & CONCLUSION

Pearson Correlation heatmap



Model evaluation results

1) Log Detection Ratio:

Model	Train R ²	Test R ²
Gradient Boosting Regressor	0.9613	0.8405
Random Forest Regressor	0.5203	0.3893

2) Current Density:

Model	Train R ²	Test R ²
Gradient Boosting Regressor	0.9936	0.7636
Random Forest Regressor	0.5387	0.1939

3) Multi-output:

Model	Train R ² (With Interaction feature)	Test R ² (With Interaction feature)	Train R ² (Without Interaction feature)	Test R ² (Without Interaction feature)
Gradient Boosting Regressor	0.9959	0.4961	0.8208	0.5444
Random Forest Regressor	0.4888	0.2581	0.7507	0.2791

Conclusion:

- GBR outperforms RF because boosting sequentially corrects errors, allowing better learning of complex relationships, whereas bagging (RF) averages independent trees and may miss subtle patterns.
- Single-output models outperform multi-output models likely due to reduced complexity and better specialization for each target variable.
- Interaction features provide limited improvement, indicating dominant influence of primary features.
- Implication: Functional groups significantly affect sensor performance, and ML models can guide optimized biosensor design for improved sensitivity and selectivity.

Literature referred :

- 1)Ciou, J. H., Li, S., & Lee, P. S. (2019). Ti₃C₂ MXene paper for the effective adsorption and controllable release of aroma molecules. Small, 15(38), 1903281.
- 2) Krishna, S., & Mir, A. (2024). Unlocking the potential of Ti₃C₂ electrodes: A data-driven capacitance prediction study. Energy Advances, 3(11), 2986–2998.

Data Spreadsheet link : https://docs.google.com/spreadsheets/d/1lW9ogmOqdy-5uW26EyPAJxs_Yq0Q-hhAaXq46BRLT-4/edit?gid=1042582767#gid=1042582767

Acknowledgement: I sincerely thank BITS-Pilani Hyderabad Campus and Afkham ma'am for helping me carrying out this project work.